



## F\*\*k you NVIDIA

Linus Torvalds, the founder & the chief developer of linux is now in the news for having said "F\*\*k you" to NVIDIA.

receivers. While RF devices do not need line-of-sight (LOS) to operate, they do perform better if they have LOS with each other. This is especially true of the 5 GHz band, which loses out on penetrative power. Also, we placed one of our test receivers (Endpoint 6), between 2 printers and a table fan, about 40 feet from the router and as expected, the signal strength and performance of that end-point was barely passable. Most devices failed to connect to it despite having the best receiver in our tests. End-point (EP) 5 was placed directly opposite EP 6, without any large, metallic objects around it, and performed quite a bit better than latter device. Another important point to note is that EP 4, which had the same wireless adapter as EPs 1 and 2, fared quite badly despite being positioned just a few feet further in Zone 2. EP 3 handled the same data far better than EP 4 due to the better receiver it housed.

## SINGLE BAND ROUTERS

These are routers capable of transmitting and receiving in only the 2.4 GHz band. We had 12 routers in this category, with prices ranging from ₹1,100 to ₹6,500. Most of these routers supported upto 300Mbps and proved themselves quite capable in our tests, some of them, such as the TP-Link TL-WR841N, even giving the dual-band routers a run for their money.

## FEATURES AND SPECIFICATIONS

Speaking in terms of design, the ASUS routers had a funky black theme, which was very similar to the nokia "Prism" series of phones which were popular some years ago. Among these routers, the ASUS RT-N15U

## HOW WE TESTED

We got a range of devices for testing, ranging from 150N routers, all the way upto 900N capable devices. To load these devices as much as possible we decided to setup 6 end-points each capable of a minimum of 150 Mbps wireless connectivity. The test setup is as follows:

**Server:** Intel Core i5- 2500k, ASUS P8Z77 V-PRO, Kingston Hyper X SSD- 240GB, 2GB RAM

**Endpoint 1:** Intel Core i7- 875k, 2GB RAM, D-Link DWA-525 N150 wireless adapter.

**Endpoint 2:** Intel QX9650, 2GB RAM, D-Link DWA-525 N150 wireless adapter.

**Endpoint 3:** Intel Q6600, 4GB RAM, D-Link DWA-525 N150 wireless adapter.

**Endpoint 4:** Dell inspiron 13R, Intel Core i5- 2430M, 2GB RAM, Intel Centrino n 1030 adapter.

**Endpoint 5:** Dell XPS 14Z, Intel Core i5- 2430M, 4GB RAM, Intel Centrino Advanced n6230 adapter.

**Endpoint 6:** Dell Alienware M14x, Intel Core i7- 2720M, 6GB RAM, Intel Centrino Advanced 6205 adapter.

All PCs were setup to run Windows 7 Home Premium and Ultimate. The TamoSoft throughput test was used to measure the throughput on all devices. The server was connected directly to the router via an ethernet cable and each router was configured to run only in 2.4 GHz mode (where applicable), at it's highest supported throughput. Each laptop was configured to run in high-performance mode, with power saving off and PCI Express link state management turned off. All other network connections were disabled on every PC, while it was being tested. The routers were reset to their factory defaults, where required. No other settings were modified other than the SSID, WPA2 passcode and LAN settings.

was the only one with internal antennas but considering the price, it's only worth going for if you really love the design and/or need the USB ports on offer. USB with UPnP support was found on 3 of the routers in our tests, but came with an exorbitant price tag, especially so in the case of the ASUS routers (RT-N16 and RT-N15U). The Buffalo WCR-HP-G300 was another unique router featuring probably the largest and

most powerful antenna (8dBi) that we had in our tests.

Coming to the interface design, all the routers in this category have simple and functional interfaces. We found the interface used by ASUS to be quite informative using a simple graphical representation to show connected devices. Essential features such as wireless SSID, etc. were also present on the home screen itself. TP-Link's interface was also

simple and clean, some might say overly simplistic, but functional nonetheless. Another important point to note is that the TP-Link and ASUS routers were the fastest to setup and connect. It isn't that the other routers are bad, it's just that while the other routers may have taken 2-4 minutes before they were detected, the ASUS and TP-Link devices responded almost immediately. This is a very minor point though, and

### Full load testing

For this test, all the PCs were placed within a 10 foot radius of the test router, TamoSoft servers were run on each of them and the network was put under simultaneous load from all six PCs. The average up and down throughput speeds were noted after five minutes.

### The remaining tests

For these tests, the end-points were placed in different zones, classified as follows.

**Zone 1:** Endpoints 1 and 2 within 10 feet of the router

**Zone 2:** Endpoints 3 and 4 between 10 and 20 feet from the router

**Zone 3:** Endpoints 5 and 6 at around 40 feet from the router.

### Distributed load tests

The TamoSoft throughput test was again conducted on all the end-points and average upload and download speeds were noted.

### Real world tests

**File transfer:** This test were conducted using a single 100MB file and around a 1000 assorted files totaling a 100MB in size. Transfer times were noted.

**Video streaming:** This test were conducted using a 1080p MTS file with a 26000 kbps bit rate. The time to play the clip was noted and points were cut for every second extra over the actual length of the file. This bit-rate is extremely high, but a router that can smoothly transmit such a signal can easily manage to stream video directly from most portable video recorders.

**Ping:** The average of 20 ping intervals were noted. Higher points were given for lower pings.

**Signal-strength:** This was noted using the PassMark WirelessMon utility. A higher-weightage was given to signal strength in farther zones.